Chapter 9

Isomorphic Graphs[4]

9.1 Graph Isomorphism

In the following three graphs:

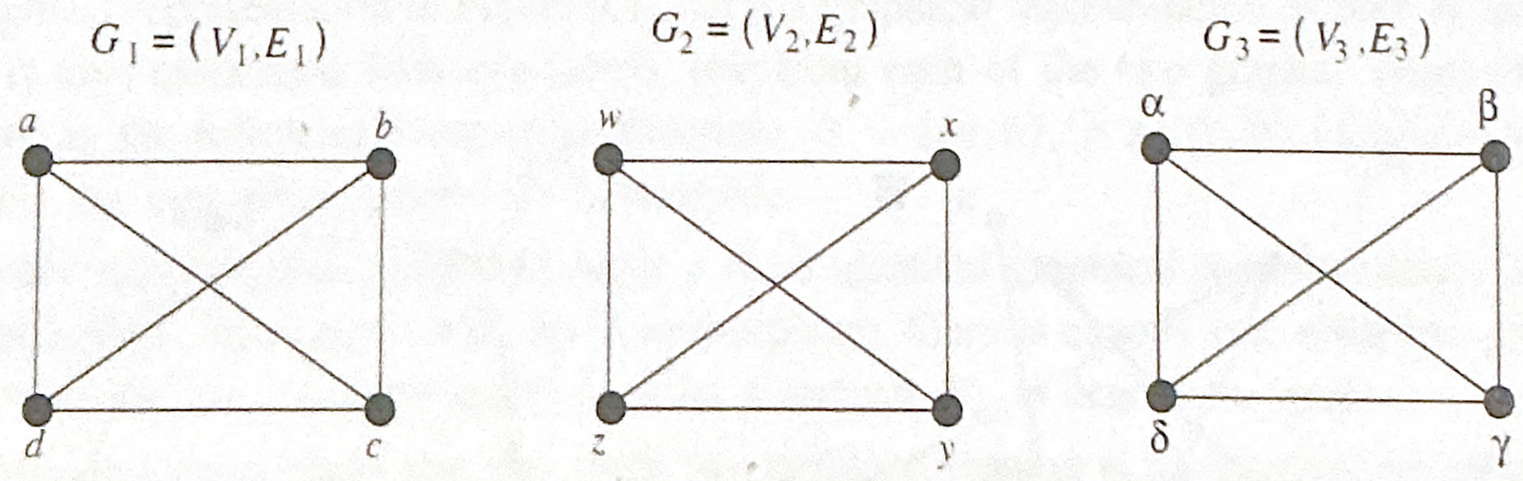
$G_1 = (V_1, E_1)$ where $V_1 = \{a, b, c, d\}$ and $E_1 = \{\{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}\}$;

$G_2 = (V_2, E_2)$ where $V_2 = \{w, x, y, z\}$ and $E_2 = \{\{w, x\}, \{w, y\}, \{w, z\}, \{x, y\}, \{x, z\}, \{y, z\}\}$;

$G_3 = (V_3, E_3)$ where $V_3 = \{\alpha, \beta, \gamma, \delta\}$ and $E_3 = \{\{\alpha, \beta\}, \{\alpha, \gamma\}, \{\alpha, \delta\}, \{\beta, \gamma\}, \{\beta, \delta\}, \{\gamma, \delta\}\}$

Although no two of them are identical to each other, they all have the same structure in their graphical representation as is evident in Figure 9.1.1.

Figure 9.1.1



The above observation motivates us to define the following.

Definition: Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. G_1 and G_2 are ***isomorphic***, denoted by $G_1 \tilde{=} G_2$, if there exists a *one-to-one* correspondence $\Phi : V_1 \to V_2$ such that $\{a, b\} \in E_1$ iff $\{\Phi(a), \Phi(b)\} \in E_2$. The function Φ is called an ***isomorphic function*** from G_1 to G_2.

[4] In this chapter, graph isomorphism is defined only for undirected simple graphs. However, it can be easily generalized to directed graphs and graphs with self-loops or parallel edges.

Example 9.1.1 The graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ in Figure 9.1.1 are isomorphic to each other. To verify this we define:

$\Phi : V_1 \rightarrow V_2$ such that $\Phi(a) = w, \Phi(b) = x, \Phi(c) = y, \Phi(d) = z$.

It is easily verified that Φ is an isomorphic function from G_1 to G_2.

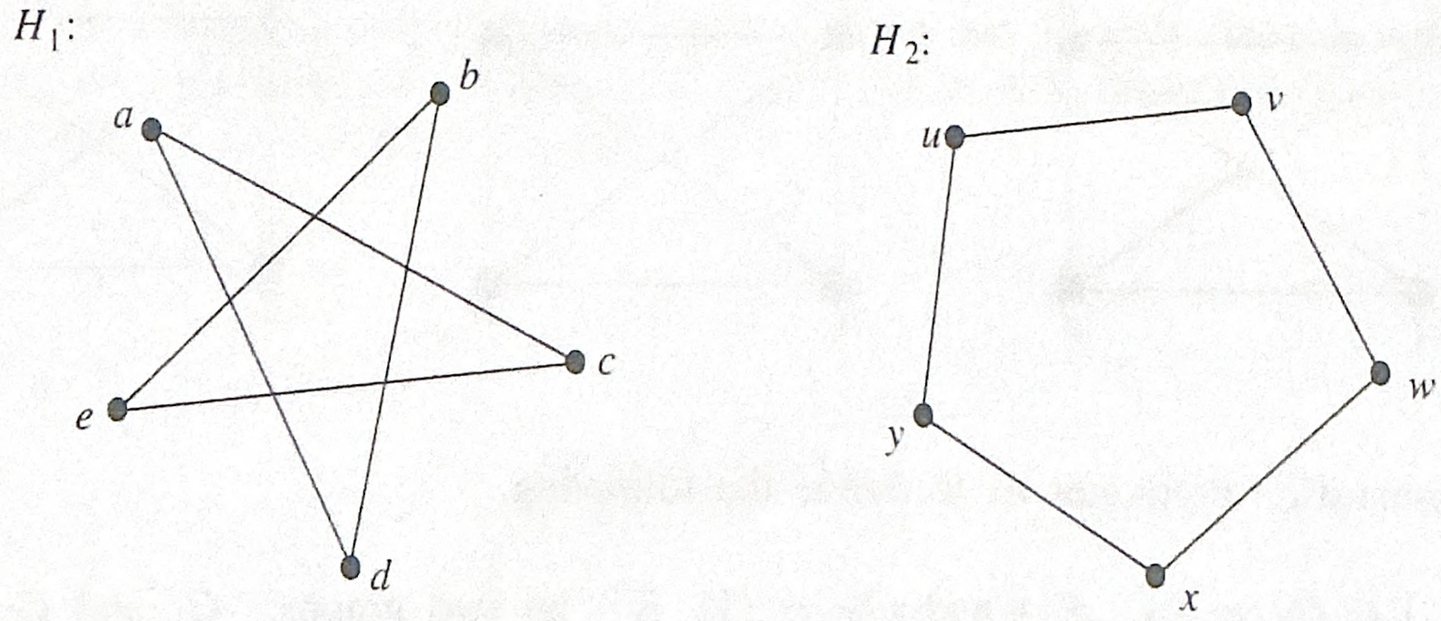
Similarly, $G_2 \underset{=}{\sim} G_3$ for if we define:

$\psi : V_2 \rightarrow V_3$ such that $\psi(w) = \alpha, \psi(x) = \beta, \psi(y) = \gamma, \psi(z) = \delta$.

Then ψ is an isomorphic function from G_2 to G_3. ■

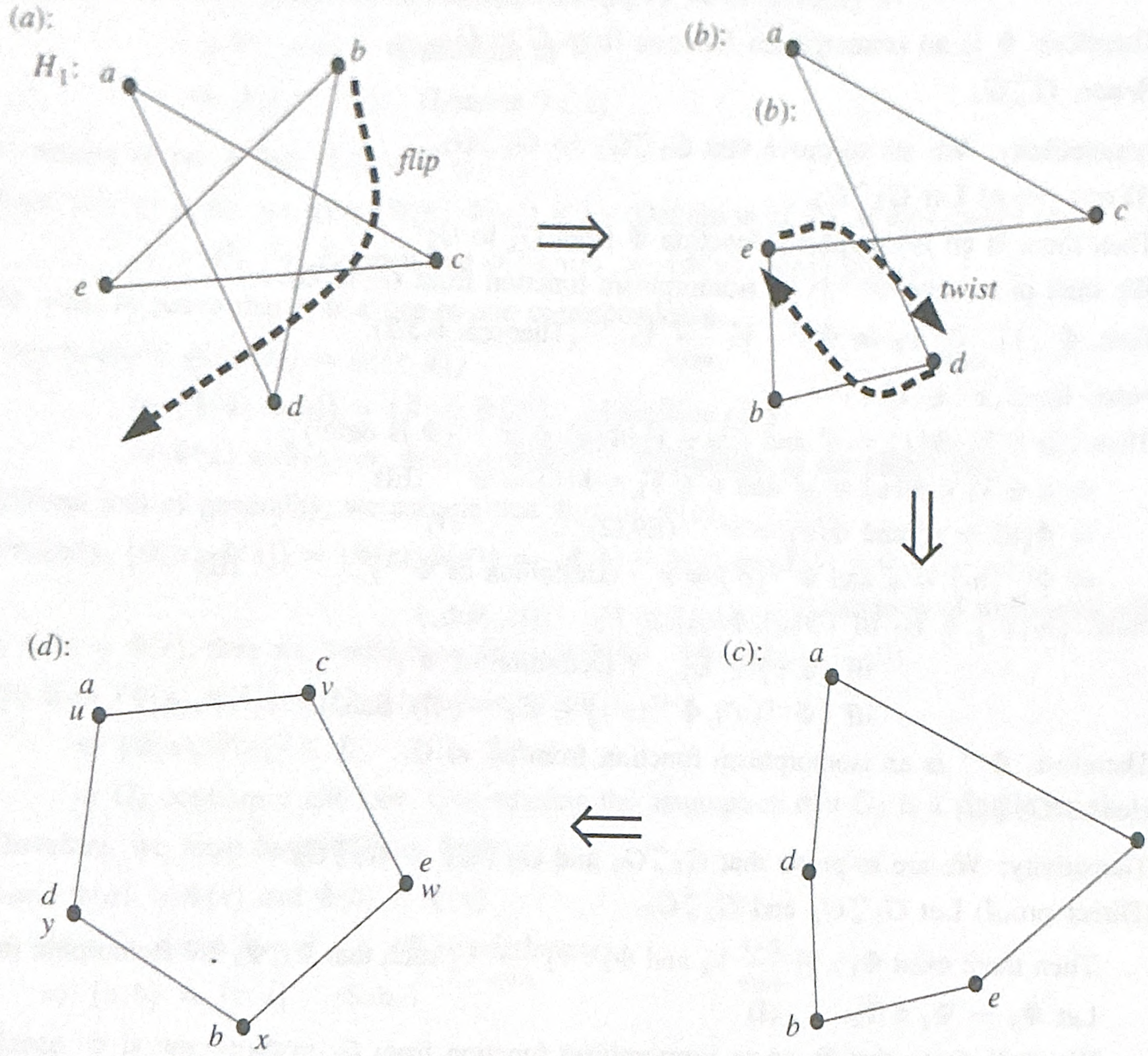
Informally speaking, two graphs are isomorphic if and only if they have identical structure, or more specifically, if and only if they have identical graphical representations (disregarding the labels). It follows that to determine if two given graphs are isomorphic, we can rely on their graphical representations as follows: first, draw a graphical representation for each graph. If the two graphical representations are identical, then the two graphs are isomorphic. Otherwise, try to transform one graphical representation into the other, or transform both into a third graphical representation. The transformation must not alter the adjacency relationship within the graphical representation. If we succeed, we can conclude that the two graphs are isomorphic. Moreover, an ordered pair can be formed from the labels of every two corresponding vertices in the two identical graphical representations, and the set of all such ordered pairs forms the desired isomorphic function.

Example 9.1.2 Prove that the following two graphs are isomorphic.



Since the graphs are already in graphical representation form, we shall proceed to compare their structures. Clearly, the two graphical representations are not identical. So, we shall try to transform one of them into the other. We choose to transform H_1 into H_2. The transformation is depicted in Figure 9.1.2. Notice that the transformation does not alter the adjacency relationship within H_1.

Figure 9.1.2



The graphical representation in Figure 9.1.2(d) is a graphical representation of both H_1 and H_2. Each vertex in it is thus associated with two labels, one from each of the two graphs. From these pairs of labels, we obtain the following isomorphic function: $\Phi = \{(a,u),(b,x),(c,v),(d,y),(e,w)\}$ from H_1 to H_2. Hence, the two given graphs are isomorphic. ■

Since every two complete graphs of order p have identical graphical representations, every two of them are isomorphic. This means that, up to isomorphism, there is exactly one complete graph of order p. We shall thus say *the* complete graph of order p and use K_p to denote *the* graph.

The following lemma states that the graph isomorphism relation is an equivalence relation.

Lemma 9.1.1: *The relation* $\underset{=}{\sim}$ *is an equivalence relation in the set of all graphs.*

Proof: (i) Reflexivity: We are to prove that $(\forall G)\big(G \text{ a graph} \Rightarrow G \underset{=}{\sim} G\big)$

(Direct proof) Let $G = (V, E)$ be a graph.

Let $\Phi = I_V$ (the identity function in V).

Then $\Phi : V \underset{onto}{\overset{1-1}{\rightarrow}} V$ such that $\Phi(v) = v, \forall v \in V. \quad \cdots$ (I)

Let $\{a, b\} \in E$. Then $a, b \in V$ (Definition of E)

$\Rightarrow \Phi(a) = a$ and $\Phi(b) = b$ ..... (II) ((I),UI)

$\Rightarrow \{a,b\} \in E \Leftrightarrow \{\Phi(a), \Phi(b)\} \in E$ ((II),Sub$_=$)

Therefore, Φ is an isomorphism function from G to G.

Hence, $G \tilde{=} G$.

(ii) Symmetricity: We are to prove that $G_1 \tilde{=} G_2 \Rightarrow G_2 \tilde{=} G_1$.

(Direct proof) Let $G_1 \tilde{=} G_2$.

Then there is an isomorphism function Φ from G_1 to G_2.

We shall prove that Φ^{-1} is an isomorphism function from G_2 to G_1.

First, $\Phi : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2 \Rightarrow \Phi^{-1} : V_2 \underset{onto}{\overset{1-1}{\rightarrow}} V_1$. (Theorem 6.3.2).

Next, let $u', v' \in V_2$.

Then $(\exists x \in V_1)\Phi(x) = u'$ and $(\exists y \in V_1)\Phi(y) = v'$ (Φ is onto.)

$\Rightarrow u \in V_1 \wedge \Phi(u) = u'$ and $v \in V_1 \wedge \Phi(v) = v'$ (EI)

$\Rightarrow \Phi(u) = u'$ and $\Phi(v) = v'$ (E9,I2) . . . . . (I)

$\Rightarrow \Phi^{-1}(u') = u$ and $\Phi^{-1}(v') = v$ (Definition of Φ^{-1}) . . . . . (II)

Now, $\{u', v'\} \in E_2$ iff $\{\Phi(u), \Phi(v)\} \in E_2$ ((I), Sub$_=$)

iff $\{u, v\} \in E_1$ (Definition of Φ)

iff $\{\Phi^{-1}(u'), \Phi^{-1}(v')\} \in E_1$ ((II), Sub$_=$).

Therefore, Φ^{-1} is an isomorphism function from G_2 to G_1.

Hence, $G_2 \tilde{=} G_1$.

(iii) Transitivity: We are to prove that $G_1 \tilde{=} G_2$ and $G_2 \tilde{=} G_3 \Rightarrow G_1 \tilde{=} G_3$.

(Direct proof) Let $G_1 \tilde{=} G_2$ and $G_2 \tilde{=} G_3$.

Then there exist $\Phi_1 : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2$ and $\Phi_2 : V_2 \underset{onto}{\overset{1-1}{\rightarrow}} V_3$ such that Φ_1, Φ_2 are isomorphic functions.

Let $\Phi_3 = \Phi_1 \circ \Phi_2$. . . . (I)

We shall prove that Φ_3 is an isomorphism function from G_1 to G_3.

First, $\Phi_1 : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2$ and $\Phi_2 : V_2 \underset{onto}{\overset{1-1}{\rightarrow}} V_3 \Rightarrow \Phi_1 \circ \Phi_2 : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_3$ (Theorem6.3.5)

$\Rightarrow \Phi_3 : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_3$. ((I), Sub$_=$)

Let $v_1, v_2 \in V_1$. Then $\{v_1, v_2\} \in E_1$

iff $\{\Phi_1(v_1), \Phi_1(v_2)\} \in E_2$ (definition of Φ_1)

iff $\{\Phi_2(\Phi_1(v_1)), \Phi_2(\Phi_1(v_2))\} \in E_3$ (definition of Φ_2)

iff $\{(\Phi_1 \circ \Phi_2)(v_1), (\Phi_1 \circ \Phi_2)(v_2)\} \in E_3$ (definition of $\Phi_1 \circ \Phi_2$)

iff $\{\Phi_3(v_1), \Phi_3(v_2)\} \in E_3$ ($\because$ (I), Sub$_=$)

Therefore, Φ_3 is an isomorphism function from G_1 to G_3.

Hence, $G_1 \tilde{=} G_3$. ■

A graph G is ***self-complementary*** if it is isomorphic to its complement (i.e. $G \tilde{=} \overline{G}$).

The following are some useful theorems which may help us in proving or disproving graph isomorphism.

Theorem 9.1.2: *Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two simple graphs such that $G_1 \tilde{=} G_2$.*

Then $|V_1| = |V_2|$ and $|E_1| = |E_2|$.

Proof: (Direct proof)

Since $G_1 \tilde{=} G_2$, there exists an isomorphic function $\Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$

$\Rightarrow V_1 \sim V_2$ (Definition of $\sim$)

$\Rightarrow |V_1| = |V_2|$ (Lemma 7.2.2)

It remains to prove that $|E_1| = |E_2|$.

Since $\forall \{u, v\} \in E_1$, we have $\{\Phi(u), \Phi(v)\} \in E_2$ (Definition of Φ). It thus makes sense to define:

$\phi : E_1 \rightarrow E_2$ such that $\phi(\{u, v\}) = \{\Phi(u), \Phi(v)\}, \forall \{u, v\} \in E_1$

We want to prove that ϕ is a one-to-one correspondence.

(ϕ is *one-to-one*:) $\phi(\{a, b\}) = \phi(\{c, d\})$

$\Rightarrow \{\Phi(a), \Phi(b)\} = \{\Phi(c), \Phi(d)\}$ (definition of ϕ)

$\Rightarrow \Phi(a) = \Phi(c)$ *or* $\Phi(a) = \Phi(d)$ (Definition of unordered pair)

Without loss of generality, we assume that $\Phi(a) = \Phi(c)$. $\cdots$ (I)

Similarly, $\{\Phi(a), \Phi(b)\} = \{\Phi(c), \Phi(d)\} \Rightarrow (\Phi(b) = \Phi(c)$ *or* $\Phi(b) = \Phi(d))$.

(Definition of unordered pair)

If $\Phi(b) = \Phi(c)$, then we would have $\Phi(a) = \Phi(b)$ ((I) $\text{Sub}_=$) $\cdots$ (II)

But then $\{\Phi(a), \Phi(b)\} \in E_2$ (Definition of ϕ)

$\Rightarrow \{\Phi(a), \Phi(a)\} \in E_2$ ((II), $\text{Sub}_=$)

$\Rightarrow G_2$ contains a self-loop, contradicting the assumption that G_2 is a simple graph.

Therefore, we must have $\Phi(b) = \Phi(d)$.

Now, $\Phi(a) = \Phi(c)$ and $\Phi(b) = \Phi(d)$

$\Rightarrow a = c$ and $b = d$ (Φ is *one-to-one*)

$\Rightarrow \{a, b\} = \{c, d\}$ ($\text{Sub}_=$)

Hence, ϕ is *one-to-one*.

(ϕ is *onto*:) Let $\{x, y\} \in E_2$. . . . (II)

Then $x, y \in V_2$. (Definition of E_2)

$\Rightarrow \exists s, t \in V_1$ such that $\Phi(s) = x$ *and* $\Phi(t) = y$ (Φ is onto)

$\Rightarrow \Phi(u) = x$ *and* $\Phi(v) = y$ (EI) . . . (III)

$\Rightarrow \{\Phi(u), \Phi(v)\} \in E_2$ ((II),(III), $\text{Sub}_=$)

$\Rightarrow \{u, v\} \in E_1$ (Definition of Φ)

It follows that $\phi(\{u, v\}) = \{\Phi(u), \Phi(v)\}$ (Definition of ϕ)

$= \{x, y\}$ ((III), $\text{Sub}_=$)

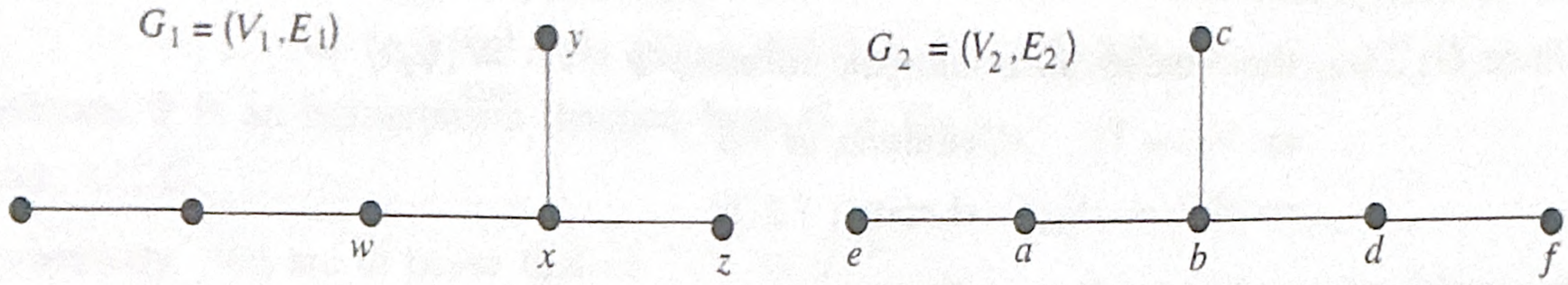
Hence, ϕ is *onto*.

Since $\phi : E_1 \overset{1-1}{\underset{onto}{\rightarrow}} E_2$, $E_1 \sim E_2$ (Definition of $\sim$)

$\Rightarrow |E_1| = |E_2|$. (Lemma 7.2.2) ■

[**Remark:** The converse of Theorem 9.1.2 is false as is shown below.

(*Proof by Counterexample*:) Consider the following two graphs.



It is obvious that $|V_1| = |V_2| = 6$ and $|E_1| = |E_2| = 5$. We shall prove that $G_1 \ncong G_2$.

(Proof by contradiction)

Suppose $G_1 \tilde{=} G_2$. Let Φ be an isomorphic function from G_1 to G_2.

Vertex x is the only vertex in G_1 that is of degree 3, and the three edges incident on x are $\{x, w\}, \{x, y\}, \{x, z\}$.

Then $\{x, w\}, \{x, y\}, \{x, z\} \in E_1$

$\Leftrightarrow \{\Phi(x), \Phi(w)\}, \{\Phi(x), \Phi(y)\}, \{\Phi(x), \Phi(z)\} \in E_2$ (Definition of isomorphic function)

$\Rightarrow deg_{G_2}(\Phi(x)) = 3.$

Since b is the only vertex of degree 3 in G_2. Therefore, $\Phi(x) = b$. $\cdots\cdots$ (I)

Next, $\{x, y\} \in E_1 \Rightarrow \{\Phi(x), \Phi(y)\} \in E_2$ (Definition of isomorphic function)

$\Rightarrow \{b, \Phi(y)\} \in E_2$ ((I), $\text{Sub}_=$)

$\Rightarrow \Phi(y) \in \{a, c, d\}$ ($\{b, a\}, \{b, c\}, \{b, d\}$ are the three edges incident on b)

If $\Phi(y) = a$, then $\{a, e\} \in E_2 \Rightarrow \{\Phi(y), e\} \in E_2$ ($\text{Sub}_=$)

$\Rightarrow \{\Phi^{-1}(\Phi(y)), \Phi^{-1}(e)\} \in E_1$ (Definition of Φ)

$\Rightarrow \{(\Phi \circ \Phi^{-1})(y), \Phi^{-1}(e)\} \in E_1$ (Definition of $\circ$)

$\Rightarrow \{y, \Phi^{-1}(e)\} \in E_1$ ($(\Phi \circ \Phi^{-1})(y) = y$)

$\Rightarrow \Phi^{-1}(e) = x$ ($\{y, x\}$ is the only edge incident on y in G_1)

$\Rightarrow \Phi(\Phi^{-1}(e)) = \Phi(x)$ (Definition of Φ)

$\Rightarrow \Phi(x) = e$ ($\Phi(\Phi^{-1}(e)) = (\Phi^{-1} \circ \Phi)(e) = e$)

$\Rightarrow \Phi(x) = e \wedge \Phi(x) = b$ (I6 with (I))

$\Rightarrow e = b$, a contradiction!

Hence, $\Phi(y) \neq a$. Similarly, $\Phi(y) \neq d$. We thus have $\Phi(y) = c$.

By a similar argument, $\Phi(z) = c$.

But then $\Phi(y) = \Phi(z) \Rightarrow y = z$, a contradiction. ($\Phi$ is one-to-one)

Hence, $G_1 \ncong G_2$. ■]

The following theorem points out that every two isomorphic graphs have the same degree sequence.

Theorem 9.1.3: *Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two simple graphs such that $G_1 \tilde{=} G_2$. Then $deg(u) = deg(\Phi(u)), \forall u \in V_1$, where Φ is an isomorphic function from G_1 to G_2.*

Proof:(Direct proof) Let $u \in V_1$.

Suppose $deg(u) = k$. $\cdots$(I)

Then there exist k distinct vertices $v_1, v_2, \cdots, v_k \in V_1$ such that $\{u, v_i\} \in E_1, 1 \leq i \leq k$.

Since Φ is an isomorphic function, $\{u, v_i\} \in E_1 \Rightarrow \{\Phi(u), \Phi(v_i)\} \in E_2, 1 \leq i \leq k$.

Moreover, Φ is one-to-one $\Rightarrow (i \neq j \Rightarrow \Phi(v_i) \neq \Phi(v_j))$ for $1 \leq i, j \leq k$.

$\Rightarrow \Phi(u)$ is adjacent at least the k distinct vertices, $\Phi(v_i), 1 \leq i \leq k$, in G_2.

$\Rightarrow deg(\Phi(u)) \geq k$ (Definition of degree)

$\Rightarrow deg(\Phi(u)) \geq deg(u)$ $((\mathrm{I}), \mathrm{Sub}_{=})$ . . . . (A)

Similarly, let $deg(\Phi(u)) = l \quad \cdots(\mathrm{II})$.

Then there exist l distinct vertices $w_1, w_2, \cdots, w_l \in V_2$ such that $\{\Phi(u), w_i\} \in E_2, 1 \leq i \leq l$.

Since Φ is onto, $\exists x_i \in V_1, \Phi(x_i) = w_i, 1 \leq i \leq l. \quad \cdots(\mathrm{III})$.

Moreover, for $1 \leq i \leq l$, $\{\Phi(u), w_i\} \in E_2 \Rightarrow \{\Phi(u), \Phi(x_i)\} \in E_2$ $((\mathrm{III}), \mathrm{Sub}_{=})$

$\Rightarrow (u, x_i) \in E_1$ (Definition of Φ)

Therefore, for $1 \leq i, j \leq l$,

$w_i \neq w_j \Rightarrow \Phi(x_i) \neq \Phi(x_j)$ $((\mathrm{III}), \mathrm{Sub}_{=})$

$\Rightarrow x_i \neq x_j$ (Φ is a function)

$\Rightarrow u$ is adjacent to at least the l distinct vertices, $x_i, 1 \leq i \leq l$, in G_1.

$\Rightarrow deg(u) \geq l$ (Definition of degree)

$\Rightarrow deg(u) \geq deg(\Phi(u))$ $((\mathrm{II}), \mathrm{Sub}_{=})$ . . . . (B)

From (A) and (B) and the antisymmetricity of $\geq$, we have $deg(u) = deg(\Phi(u)), \forall u \in V_1$. ■

[Remark: The converse of Theorem 9.1.3 is false as is shown below.

(*Proof by Counterexample*:) Consider the two graphs given in the Remark after Theorem 9.1.2. The two graphs have the same degree sequence 3, 2, 2, 1, 1, 1. However, it has been shown above that they are not isomorphic.]

Since the converse of both Theorem 9.1.2 and Theorem 9.1.3 are false, the two theorems can only be used in proving that two graphs are *not* isomorphic. They cannot be used in proving that two graphs are isomorphic.

Theorem 9.1.4: *Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two simple undirected graphs.*

$$G_1 \tilde{}_{=} G_2 \quad \textit{iff} \quad \overline{G_1} \tilde{}_{=} \overline{G_2}.$$

Proof: First note that as $G_i, \overline{G_i}$, i=1,2, are simple graphs,

$\forall u, v \in V_i, (u = v) \Rightarrow (\{u, v\} \in E_i \Leftrightarrow \{u, v\} \in \overline{E_i})$ . . . . (I) (This chapter, Exercise 6)

$G_1 \tilde{}_{=} G_2$

iff $\exists \Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$ such that $\forall u, v \in V_1, (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2)$ $\left(definition\ of\ \tilde{}_{=}\right)$

iff $\exists \Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)((u \neq v) \Rightarrow (u, v \in V_1 \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2))) \wedge$

$(\forall u)(\forall v)((u = v) \Rightarrow (u, v \in V_1 \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2)))$ (Chapter 2, Exercise 2.1)

iff $\exists \Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)(u, v \in V_1 \Rightarrow ((u \neq v) \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2))) \wedge$

$(\forall u)(\forall v)(u, v \in V_1 \Rightarrow ((u = v) \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2)))$ (Chapter 1, Exercise 2.o)

iff $\exists \Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)(u, v \in V_1 \Rightarrow ((u \neq v) \Rightarrow (\{u, v\} \notin E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \notin E_2))) \wedge$

$(\forall u)(\forall v)(u, v \in V_1 \Rightarrow ((u = v) \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2)))$ (Theorem 1.7.2(iii))

iff $\exists \Phi : V_1 \overset{1-1}{\underset{onto}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)\big(u, v \in V_1 \Rightarrow \big((u \neq v) \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)\big) \wedge$

$(\forall u)(\forall v)(u, v \in V_1 \Rightarrow ((u = v) \Rightarrow (\{u, v\} \in E_1 \Leftrightarrow \{\Phi(u), \Phi(v)\} \in E_2)))$ (Definitions of $\overline{G_1}, \overline{G_2}$)

iff $\exists \Phi : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)\big(u, v \in V_1 \Rightarrow \big((u \neq v) \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)\big) \wedge$

$(\forall u)(\forall v)\big(u, v \in V_1 \Rightarrow \big((u = v) \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)\big)$ (by (I))

iff $\exists \Phi : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)\big((u \neq v) \Rightarrow \big(u, v \in V_1 \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)\big) \wedge$

$(\forall u)(\forall v)\big((u = v) \Rightarrow \big(u, v \in V_1 \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)\big)$ (Chapter 1, Exercise 2.o)

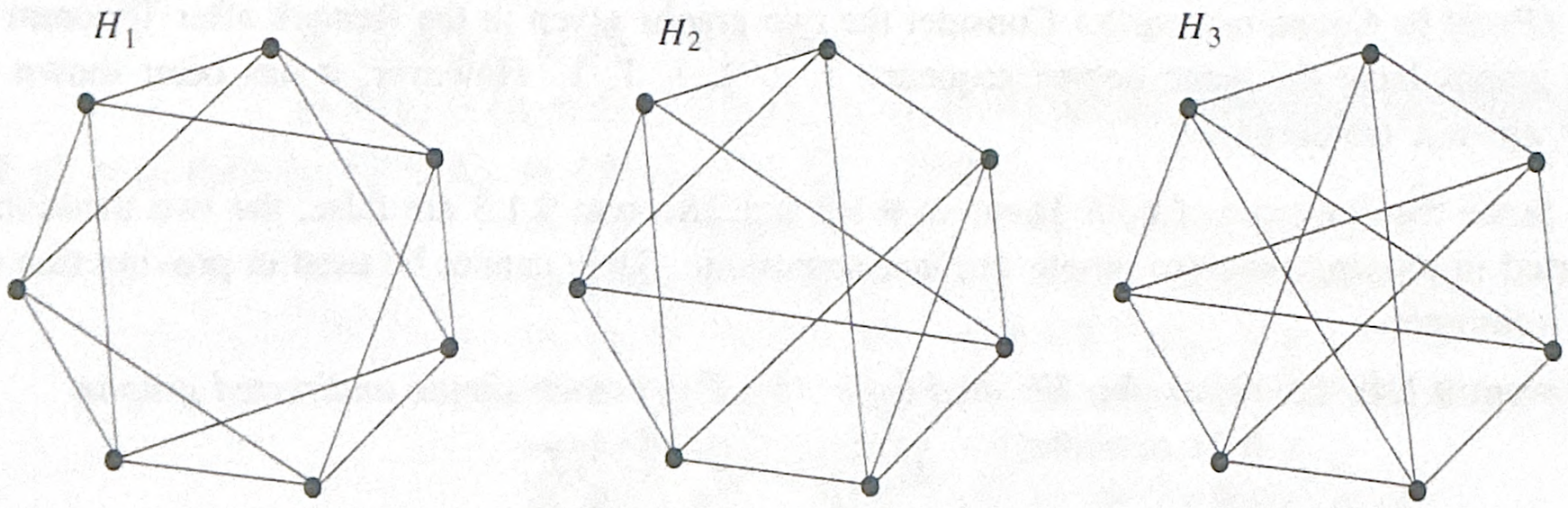
iff $\exists \Phi : V_1 \underset{onto}{\overset{1-1}{\rightarrow}} V_2$ such that

$(\forall u)(\forall v)\big(u, v \in V_1 \Rightarrow \big(\{u, v\} \in \overline{E_1} \Leftrightarrow \{\Phi(u), \Phi(v)\} \in \overline{E_2}\big)\big)$ (Chapter 2, Exercise 2.l)

iff $\overline{G_1} \tilde{=} \overline{G_2}$ (definition of $\tilde{=}$). ■

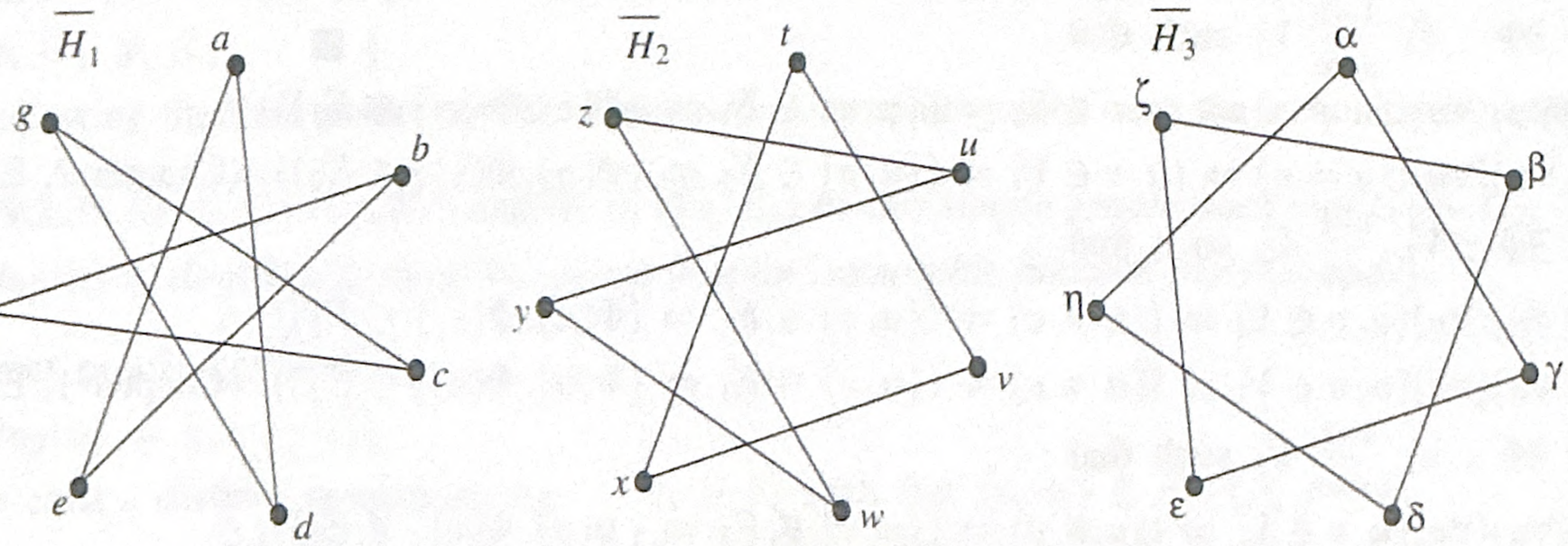
The above theorem points out that if G_1 and G_2 have more edges than $\overline{G_1}$ and $\overline{G_2}$, then in testing if $G_1 \tilde{=} G_2$, it would be easier to test if $\overline{G_1} \tilde{=} \overline{G_2}$. The following is an example.

Example 9.1.3 Which of the following graphs are isomorphic?



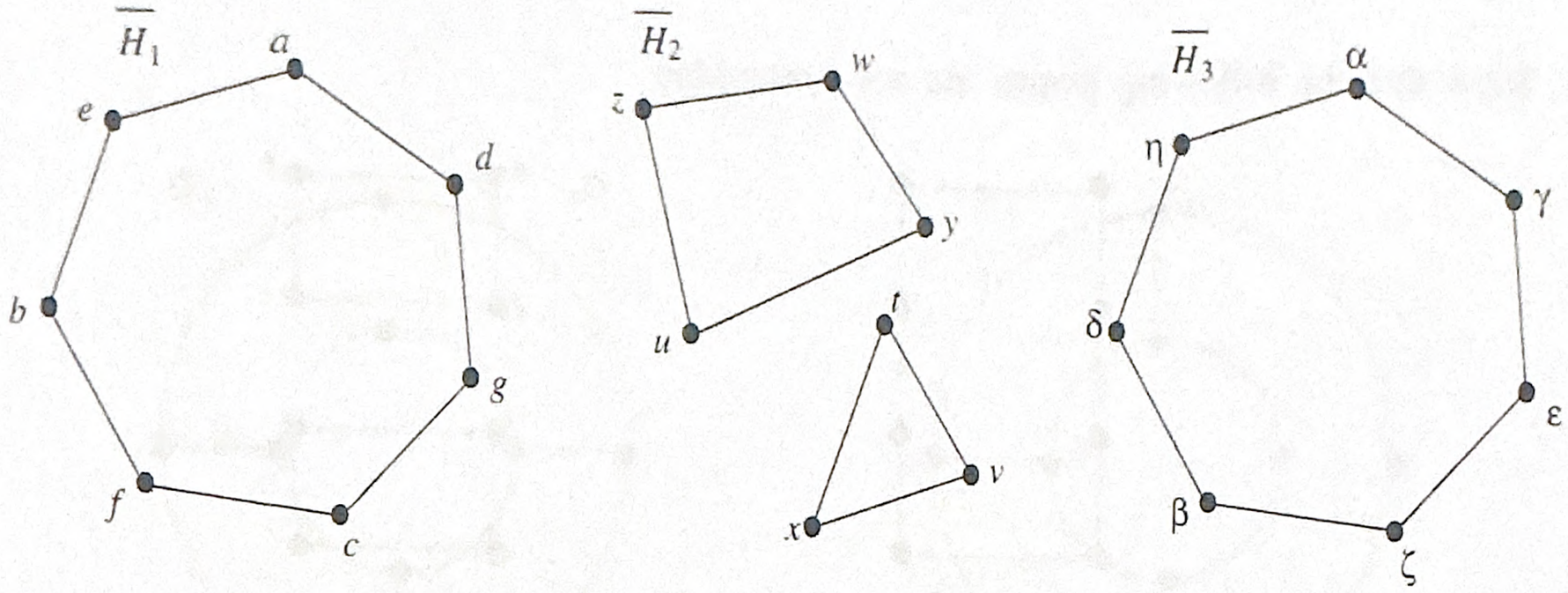
Solution: Construct the complements of H_1, H_2 and H_3. Their graphical representations are shown in Figure 9.1.3. Notice that the vertices are labelled.

Figure 9.1.3 The complements of H_1, H_2 and H_3.



The graphical representations are transformed into less obscure forms depicted in Figure 9.1.4. Since $\overline{H_1}$ and $\overline{H_3}$ have identical graphical representation, $\overline{H_1} \tilde{=} \overline{H_3}$ and the function $\Phi = \{(a, \alpha), (b, \delta), (c, \zeta), (d, \gamma), (e, \eta), (f, \beta), (g, \varepsilon)\}$ is an isomorphic function from $\overline{H_1}$ to $\overline{H_3}$.

Figure 9.1.4



On the other hand, $\overline{H_1} \not\cong \overline{H_2}$ for if they were, then there would exist an isomorphic function $\Phi' : V_{H_2} \underset{onto}{\overset{1-1}{\rightarrow}} V_{H_1}$. Since vertices t, v and x are mutually adjacent in $\overline{H_2}$, their images under Φ' in $\overline{H_1}$ must also be mutually adjacent. However, by inspection, there is no such three vertices in $\overline{H_1}$. Hence, $\overline{H_1} \not\cong \overline{H_2}$.

By Theorem 9.1.4, we have $H_1 \cong H_3$ and $H_1 \not\cong H_2$. It follows from Theorem 9.1.1 that $H_2 \not\cong H_3$. ■

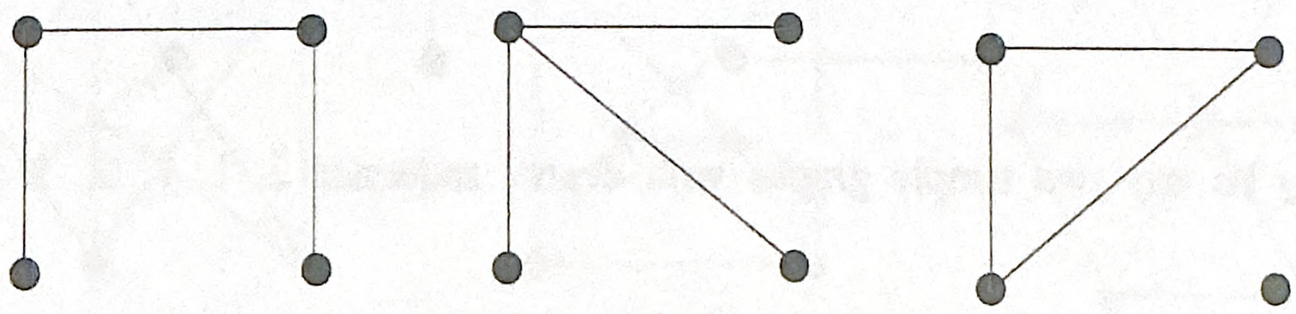
An ***invariant*** is a graph property that is preserved under isomorphism. Specifically, a property P is an *invariant* if given any two graphs G_1 and G_2 such that $G_1 \cong G_2$, G_1 has property P if and only if G_2 has property P.

Theorem 9.1.2 and Theorem 9.1.3 show that the order, size, and degree sequence are invariants. Other invariants are given in Section 10.5, Example 7 and Section 10.6, Exercise 5.

To prove that two graphs are isomorphic, we must find an isomorphism from one graph to the other. To prove that two graphs are *not* isomorphic, we may show that they differ in an invariant. This can be done by showing that one graph has a specific structure that the other does not have. The remarks after Theorem 9.1.2 and Theorem 9.1.3 and the above example illustrate the idea.

9.2 Examples

1. Show that no two of the following graphs are isomorphic.



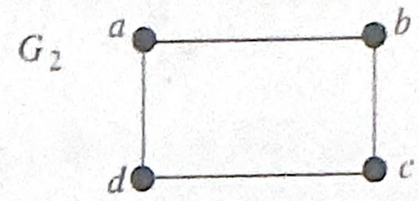
Solution: The degree sequences of the three graphs are

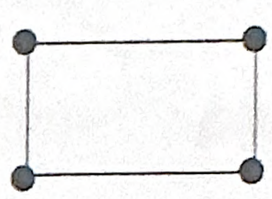
2, 2, 1, 1
3, 1, 1, 1
2, 2, 2, 0

Since no two of the degree sequences are the same, by Theorem 9.1.3, no two of the graphs are isomorphic. ■

2. Show that the following graphs are not isomorphic.







Solution: (*Proof by Contradiction*): Suppose $G_2 \tilde{=} G_1$.

Let Φ be an isomorphic function from G_2 to G_1. Then,

$\{a,b\},\{b,c\},\{c,d\},\{d,a\} \in E_{G_2}$ iff $\{\Phi(a),\Phi(b)\},\{\Phi(b),\Phi(c)\},\{\Phi(c),\Phi(d)\},\{\Phi(d),\Phi(a)\} \in E_{G_1}$

which implies that G_1 contains four edges forming a cycle. However, by inspection, G_1 has no such a cycle, a contradiction! Hence, $G_1 \not\tilde{=} G_2$. ■

3. Let G_1, G_2 and G_3 be any three simple graphs of order 4 and size 2. Prove that at least two of them are isomorphic.

Solution: Let G be a graph of order 4 and size 2. Then $|E_G| = 2 \Rightarrow \sum_{v \in V_G} deg(v) = 2|E_G| = 2 \times 2 = 4$.

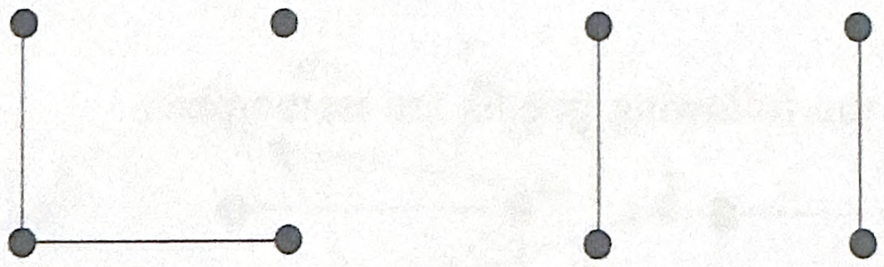
Moreover, since G is a simple graph, $|E_G| = 2 \Rightarrow deg(v) \leq 2, \forall v \in V_G$.

Therefore, the possible degree sequence for G are:

(i) 2, 2, 0, 0
(ii) 2, 1, 1, 0
(iii) 1, 1, 1, 1

Degree sequence (*i*) is impossible because G is a simple graph.

For degree sequences (*ii*) and (*iii*), G has the following graphical representations:



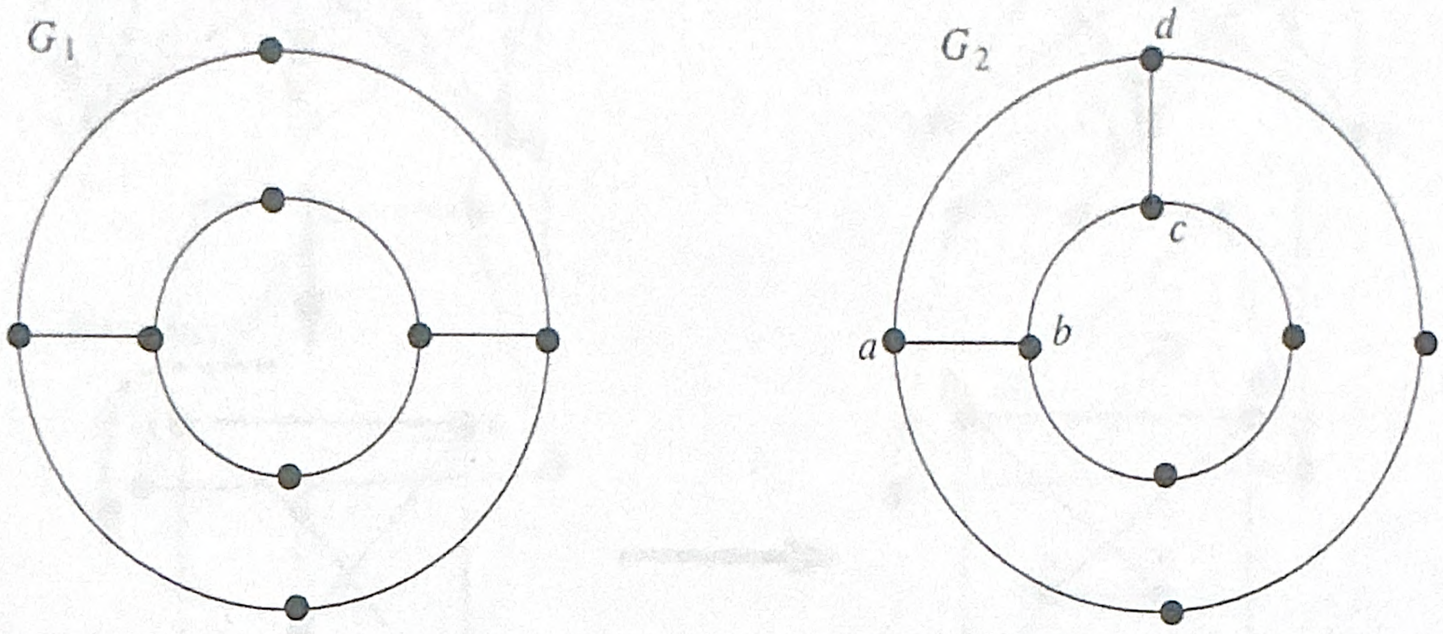
Let G_1, G_2 be any two simple graphs with degree sequence 2, 1, 1, 0. It is easily verified that $G_1 \tilde{=} G_2$.

Similarly, for any two simple graphs with degree sequence 1, 1, 1, 1, it is easily verified that they are isomorphic.

Hence, there are exactly two equivalence classes w.r.t. $\tilde{=}$ in the set of all simple graphs of order 4 and size 2.

By the pigeonhole principle, among three simple graphs of order 4 and size 2, at least $\lceil 3/2 \rceil = 2$ of them belong to the same equivalence class and hence are isomorphic. ■

4. Show that the following two graphs are not isomorphic.



Solution: (*Proof by Contradiction*)

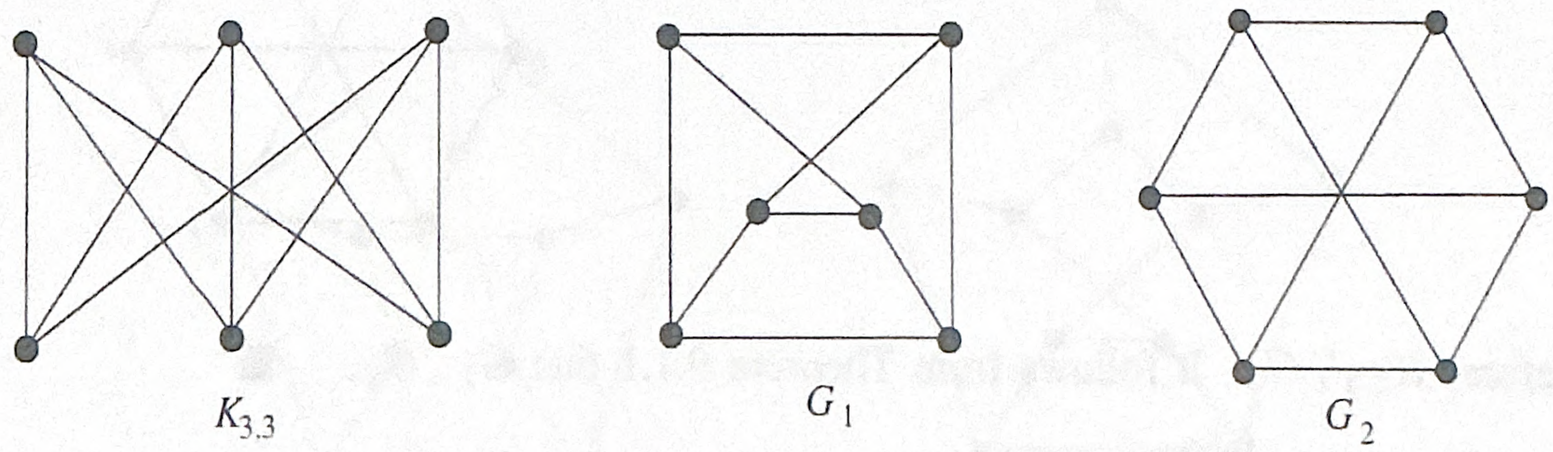
Suppose $G_1 \tilde{=} G_2$. Then there exists an isomorphic function Φ from G_2 to G_1.

Since, $a\ b\ c\ d\ a$ form a cycle of four edges in which every vertex is of degree 3 in G_2, By the definition of Φ and Theorem 9.1.3, their images $\Phi(a)\ \Phi(b)\ \Phi(c)\ \Phi(d)\ \Phi(a)$ must also form a cycle of four edges in which every vertex is of degree 3 in G_1.

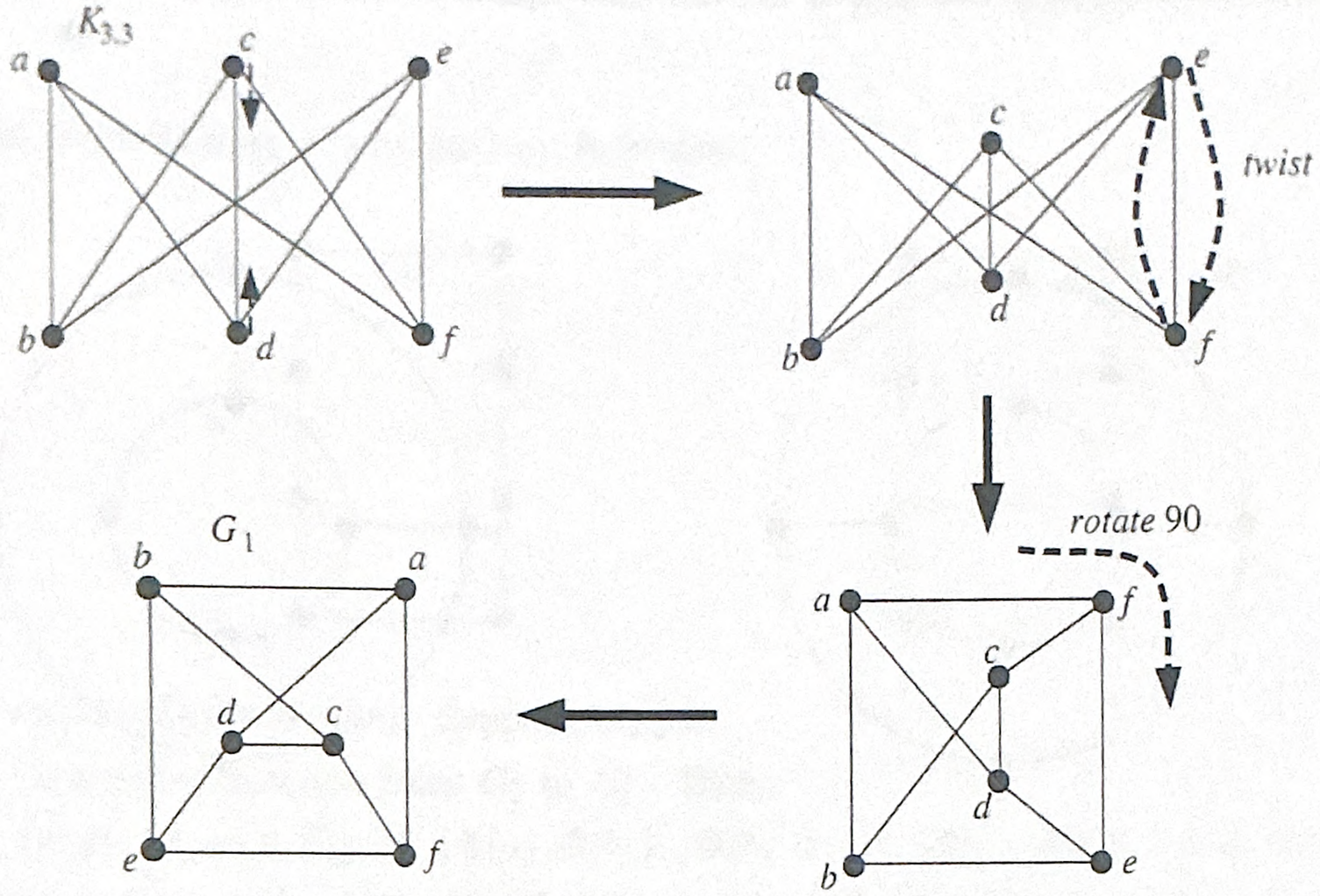
However, by inspection, G_1 has no such cycle, a contradiction.

Hence, $G_1 \not\tilde{=} G_2$. ■

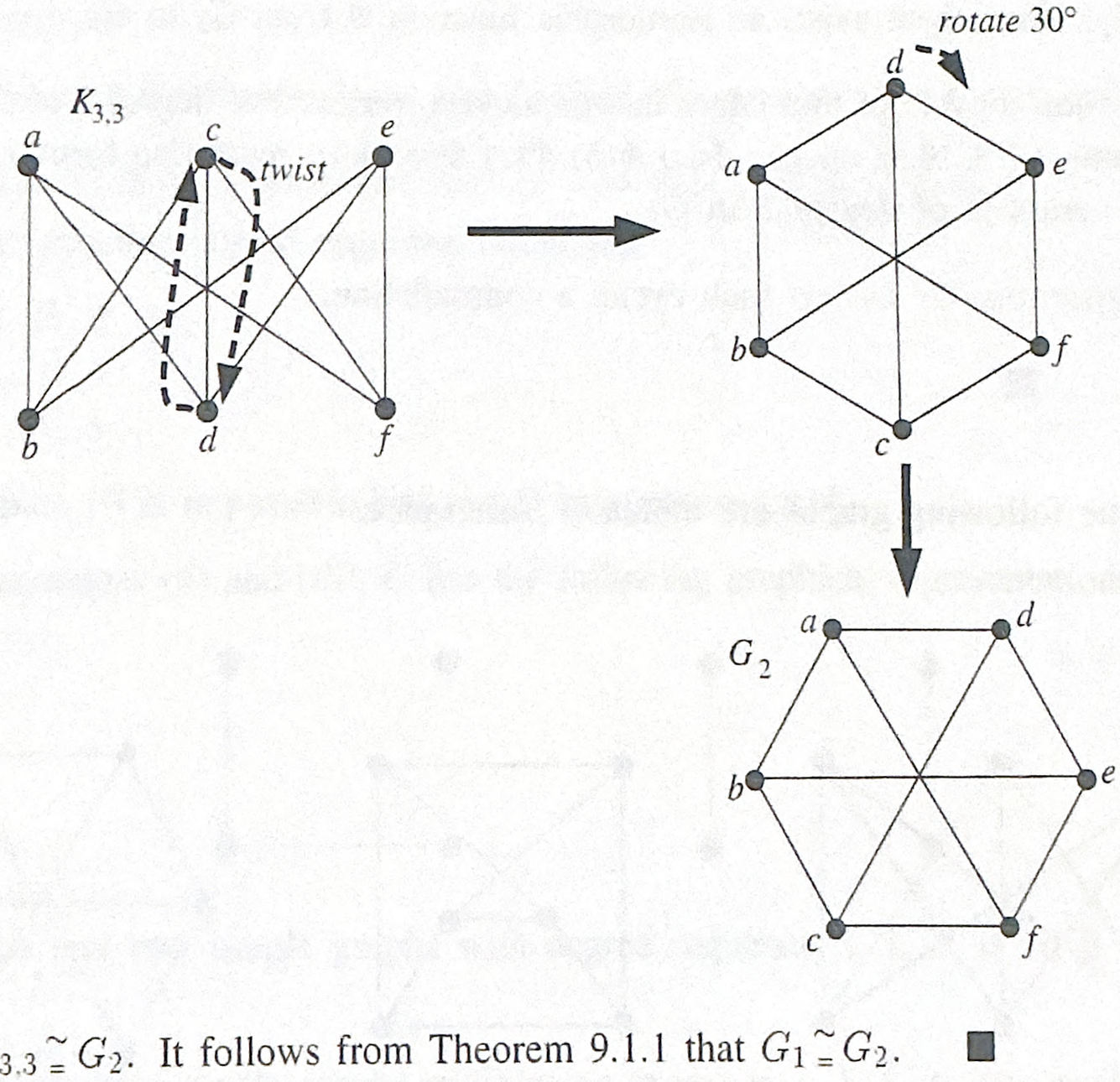
5. Show that the following graphs are mutually isomorphic.



Solution:



Therefore, $K_{3,3} \tilde{=} G_1$.

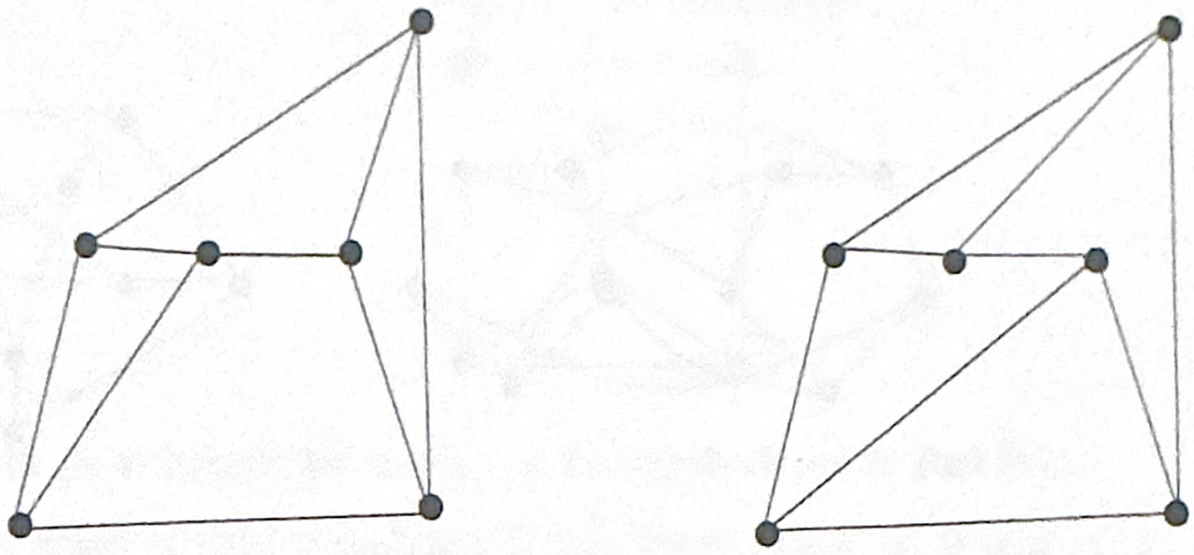


Therefore, $K_{3,3} \tilde{=} G_2$. It follows from Theorem 9.1.1 that $G_1 \tilde{=} G_2$. ■

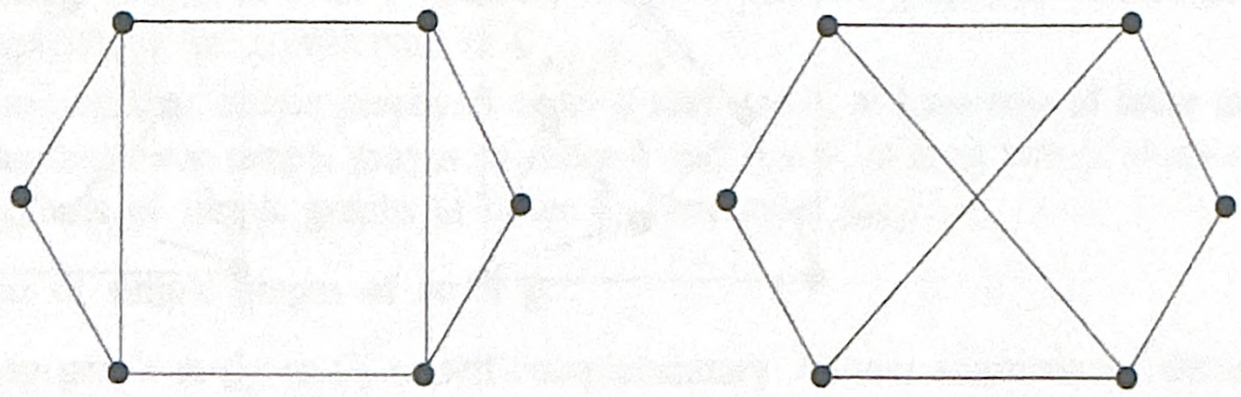
9.3 Exercises

1. Determine which of the following pairs of graphs are isomorphic.

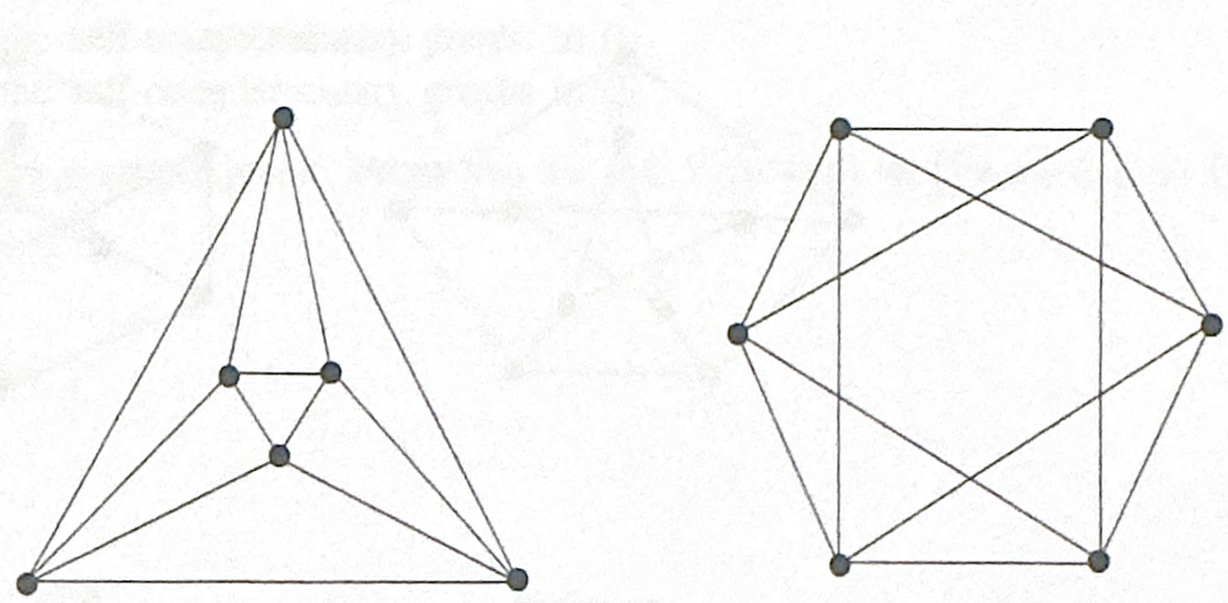
(a)



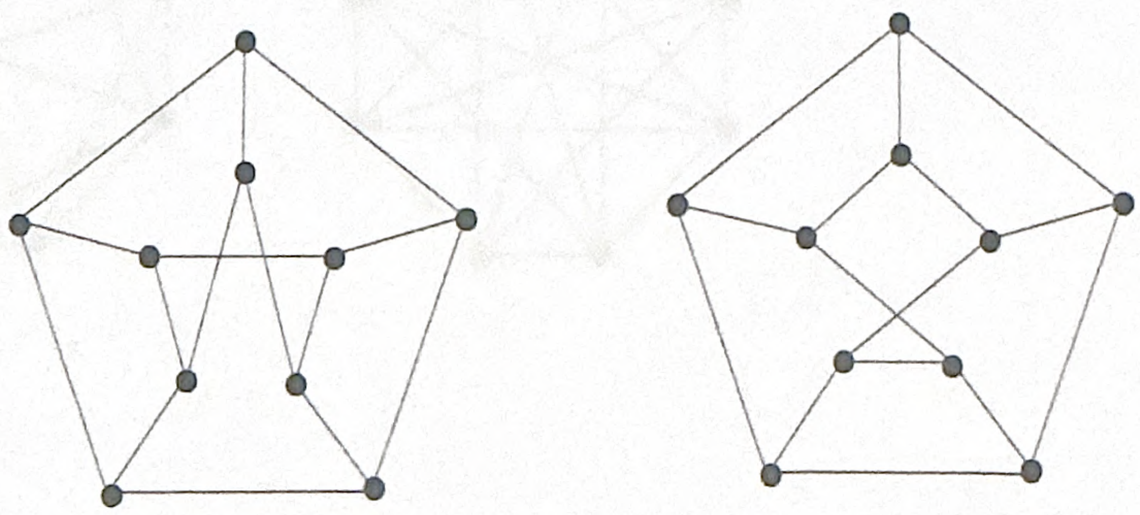
(b)



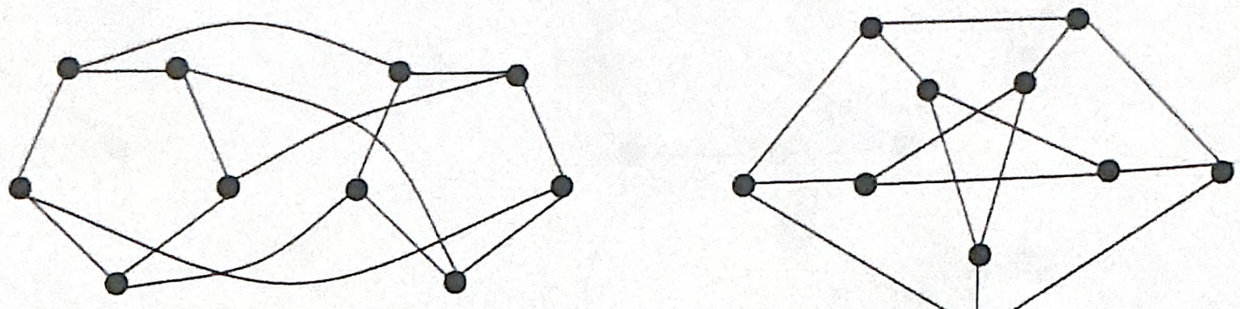
(c)



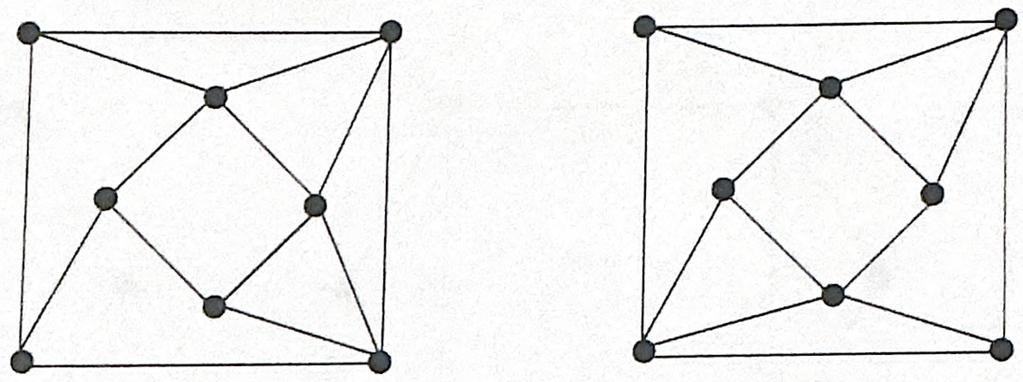
(d)



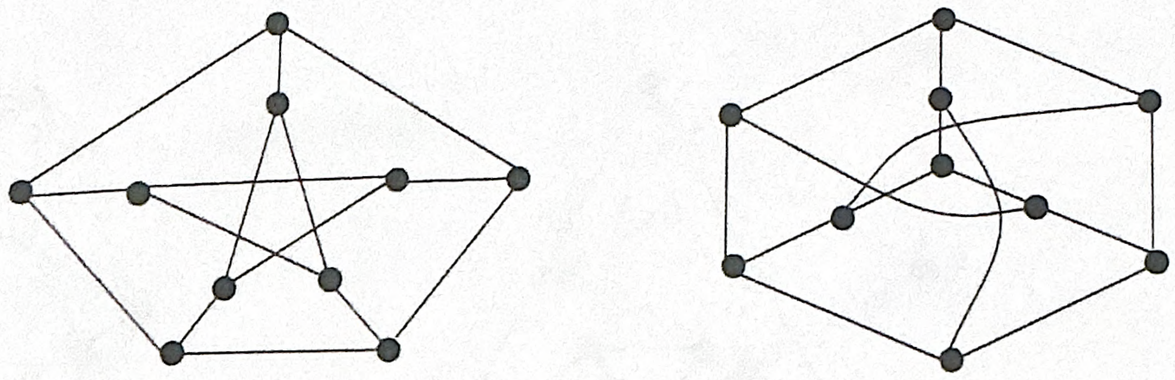
(e)



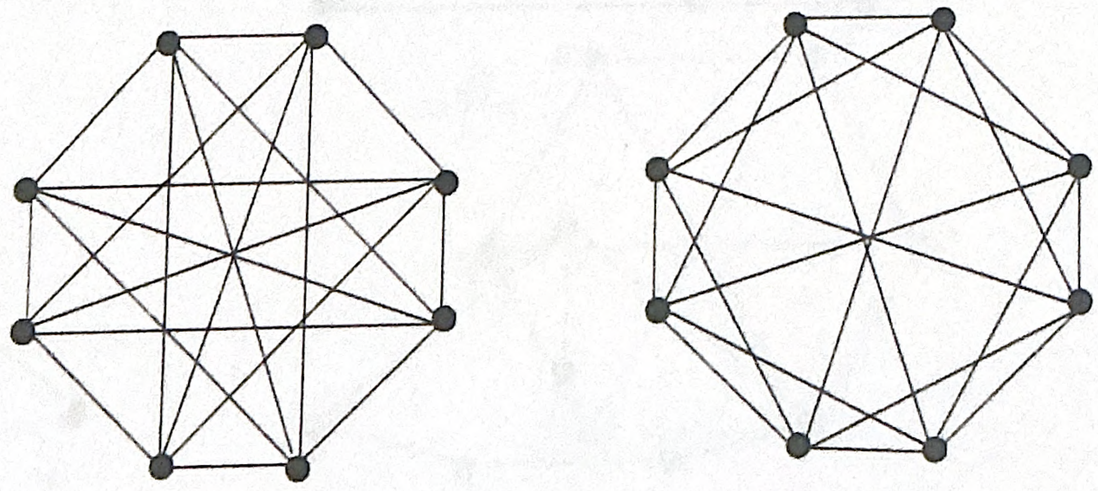
(f)



(g)

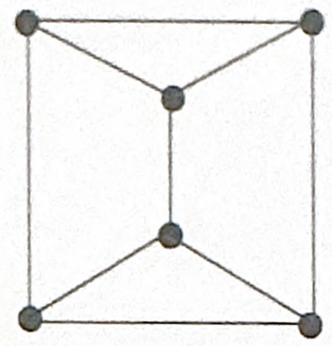


(h)



2. Let $\mathcal{G}_n$ be the set of all 3–regular simple graphs of order n. Prove that:
   (a) If $G \in \mathcal{G}_n$, then n is a multiple of 2 and $|E_G|$ is multiple of 3.
   (b) $K_{3,3}$ is *not* isomorphic to the following *prism graph*.

Figure 5 The Prism graph



   (c) Every graph in $\mathcal{G}_6$ is isomorphic to $K_{3,3}$ or the graph shown in Part (b).

3. Let G be a simple graph of order p such that $G \tilde{=} \overline{G}$. Prove that $p \equiv_4 0$ or $p \equiv_4 1$.

4. (a) Let $\mathcal{G}_p$ be the set of simple graphs of order p. For each integer q such that $0 \leq q \leq \binom{p}{2}$, let $\mathcal{G}_{p,q}$ be the set of simple graphs of order p and size q. Prove that two graphs are isomorphic in $\mathcal{G}_{p,q}$ if and only if their complements are isomorphic in $\mathcal{G}_{p,\binom{p}{2}-q}$.
   (b) Prove that among four simple graphs of order 4 and size 3, at least two of them are isomorphic.
   (c) Prove that among three simple graphs of order 4 and size 4, at least two of them are isomorphic.
   (d) Let $\mathcal{G}_4$ be the set of simple graphs of order 4. Determine $[\mathcal{G}_4]/\tilde{=}$.

5. Let $\mathcal{G}_p$ be the set of simple graphs of order p.
   (a) Prove that no graph in $\mathcal{G}_2$ or $\mathcal{G}_3$ is self-complementary without enumerating the equivalence classes in $[\mathcal{G}_2]/\tilde{=}$ or $[\mathcal{G}_3]/\tilde{=}$.
   (b) Determine the self-complementary graphs in $\mathcal{G}_4$.
   (c) Determine the self-complementary graphs in $\mathcal{G}_5$.

6. Let $G = (V, E)$ be a simple graph. Prove that $\forall u, v \in V, (u = v) \Rightarrow (\{u, v\} \in E \Leftrightarrow \{u, v\} \in \overline{E})$.

(a)

